North American Particle Accelerator Conference, NAPAC2019,
Lansing, Michigan USA, 1-6 September 2019.
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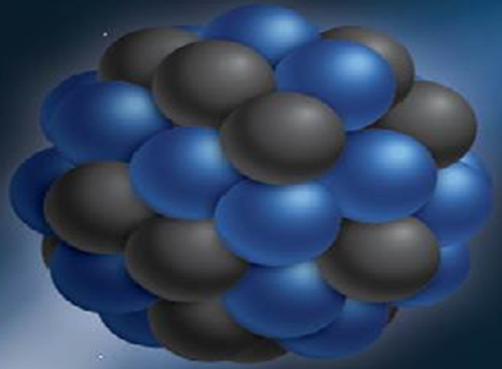
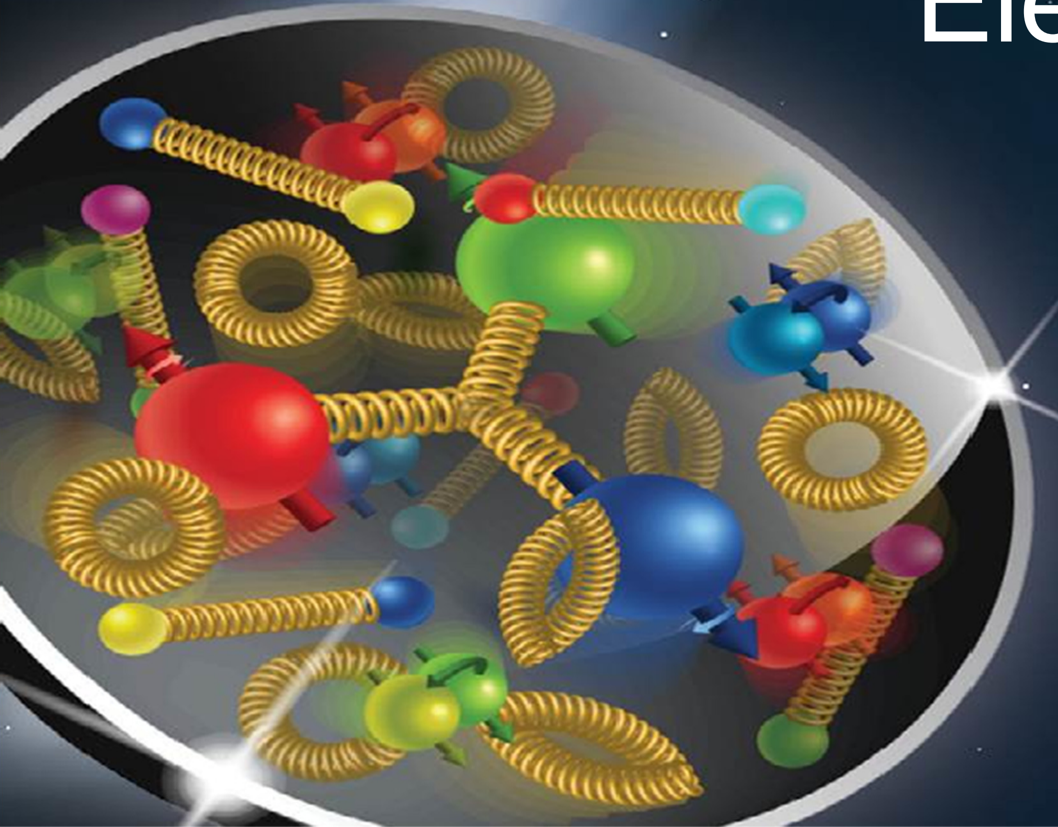




Electron Ion Collider Machine Detector Interface

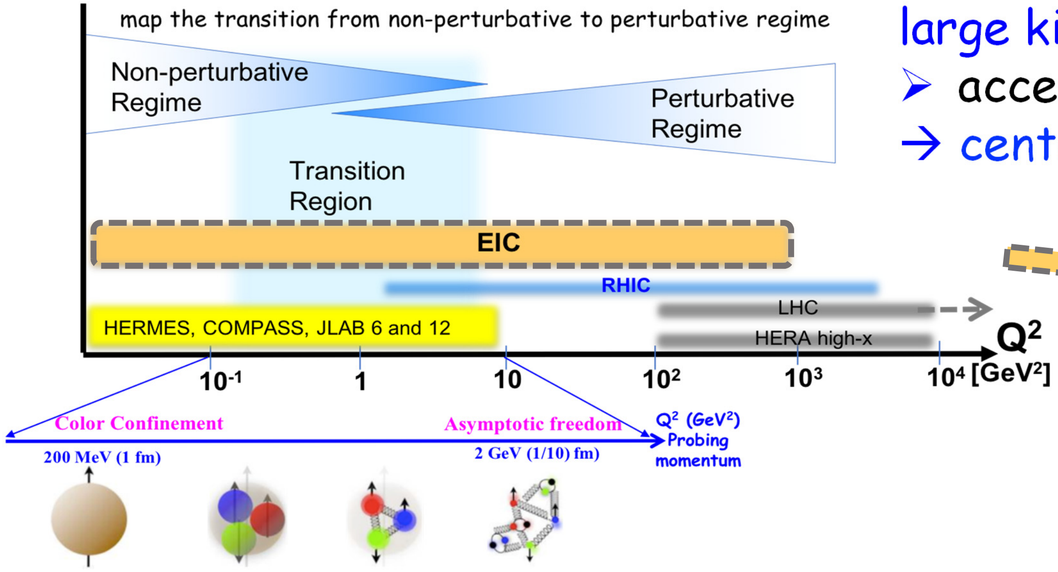
B. Parker, E.C. Aschenauer, A. Kiselev, C. Montag, R. B. Palmer, V. Ptitsyn, F. Willeke and H. Witte – BNL, M. Diefenthaler, Y. Furletova, V. Morozov, D. Romanov, T. Michalski, A. Seryi and R. Yoshida – JLab, C. Hyde – ODU, M. Sullivan – SLAC,

September 3, 2019

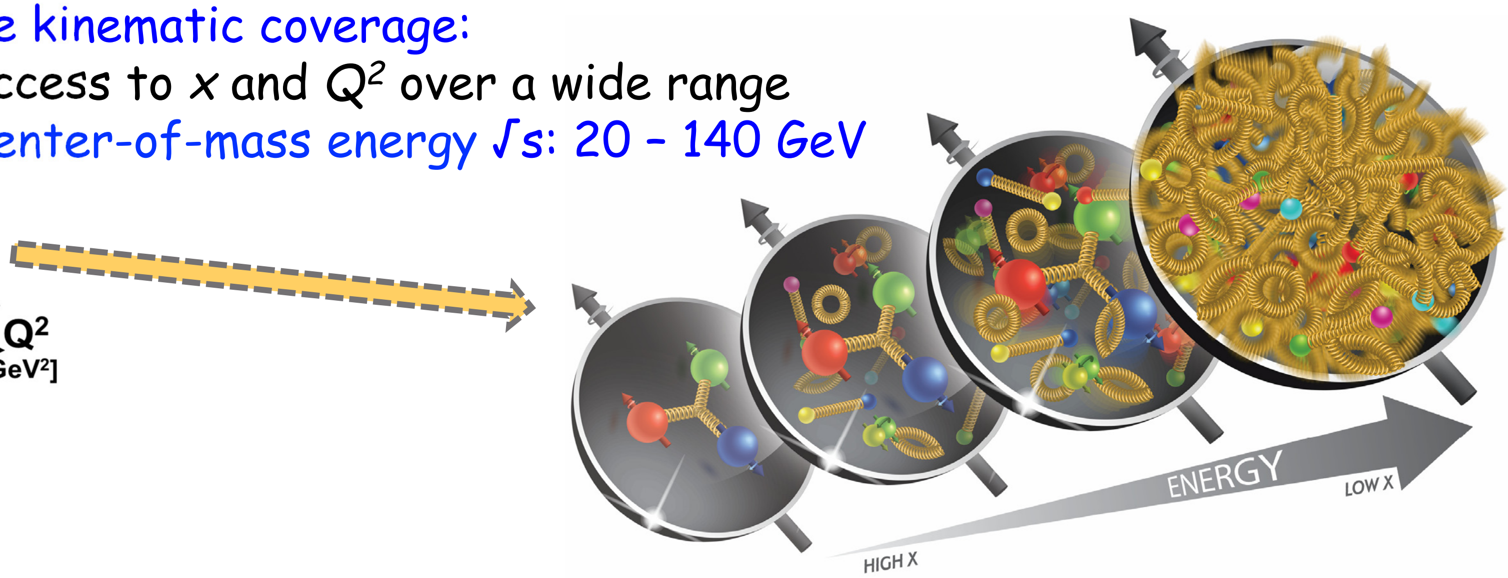
Electron Ion Collider – EIC

Jefferson Lab BROOKHAVEN NATIONAL LABORATORY U.S. DEPARTMENT OF ENERGY Office of Science

What is needed to address EIC Physics?



large kinematic coverage:
- access to x and Q^2 over a wide range
- → center-of-mass energy $\sqrt{s}$: 20 – 140 GeV



polarized electron and hadron (p,d,He-3) beams:
- access to spin structure of nucleons and nuclei
- Spin vehicle to access the spatial and momentum structure of the nucleon in 3d
- Full specification of initial and final states to probe q-g structure of NN and NNN interaction in light nuclei

nuclear beams: D to Pb
- accessing the highest gluon densities → saturation
- quark and gluon interact with a nuclear medium

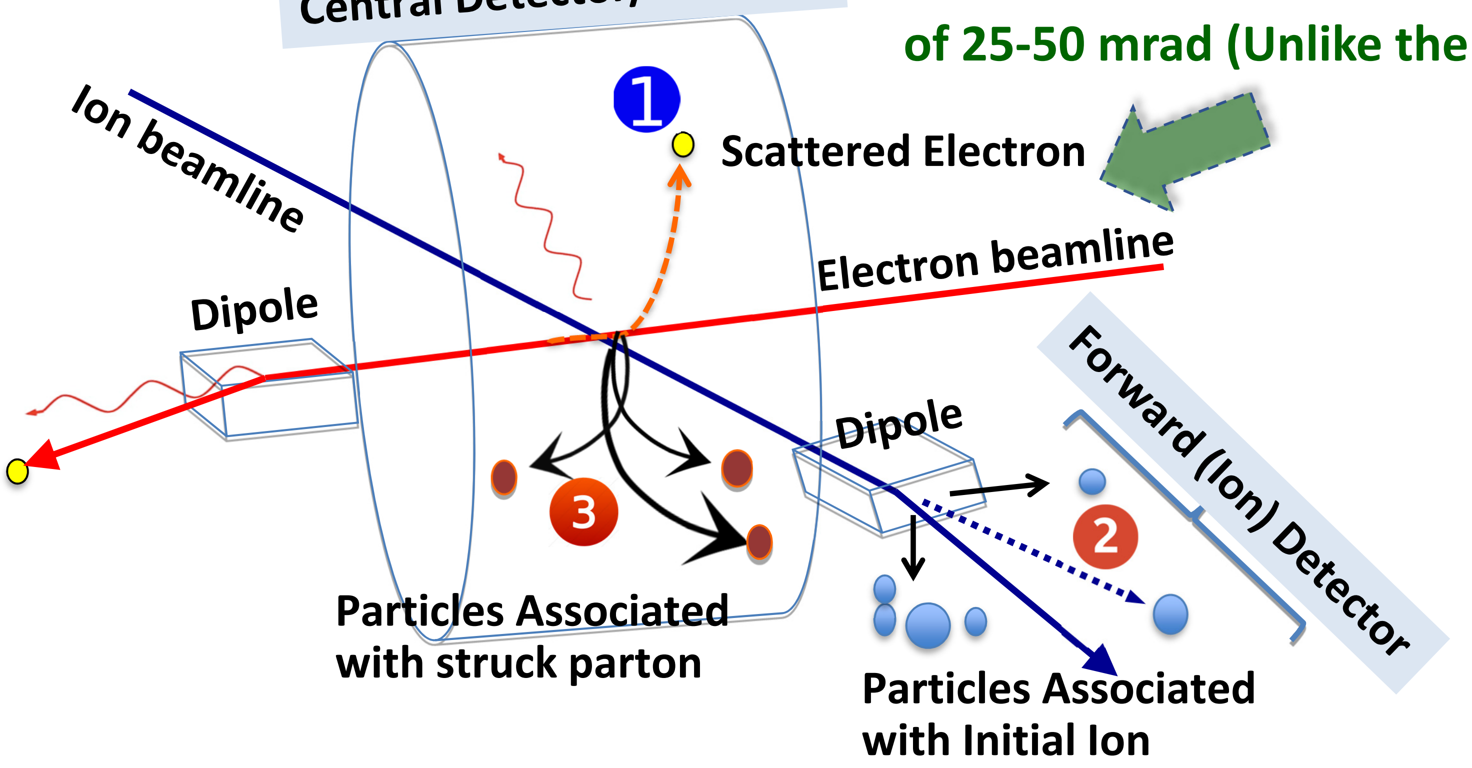
high luminosity $10^{33\text{-}34}$ cm^{-2}s^{-1}:
- mapping the spatial and momentum structure of nucleons and nuclei in 3d
- access to rare probes, i.e. Ws

EIC challenges are different from other colliders.

EIC	LHC / RHIC
Collide different beam species: ep & eA → consequences for beam backgrounds → hadron beam backgrounds, i.e. beam gas events → synchrotron radiation	Collide the same beam species: pp, pA, AA → beam backgrounds → hadron beam backgrounds, i.e. beam gas events, high pile up
Asymmetric beam energies → boosted kinematics → high activity at high $\lvert\eta\rvert$	Symmetric beam energies → kinematics is not boosted → most activity at midrapidity
<u>High repetition rate</u> <u>2 – 9 ns spacing between bunches</u>	<u>Moderate repetition rate</u> <u>25 ns between bunches</u>
Large crossing angle Crab Crossing: 25 – 50 mrad	No significant crossing angle… yet
Wide range in center of mass energies → **factor 7**	Limited range in center of mass energies → **LHC: factor 2** → **RHIC: factor 26 in AA and 8 in pp**
EIC both beams are polarized → stat uncertainty: ~ $1/(P_1 P_2 (\int \mathcal{L} dt)^{1/2})$	LHC no beam polarization (RHIC has pp) → stat uncertainty: ~$1/(\int \mathcal{L} dt)^{1/2}$

Differences impact detector acceptance and possible detector technologies!

What does an idealized EIC detector have?



Note: Our EIC has a crossing angle of 25-50 mrad (Unlike the LHeC IR)

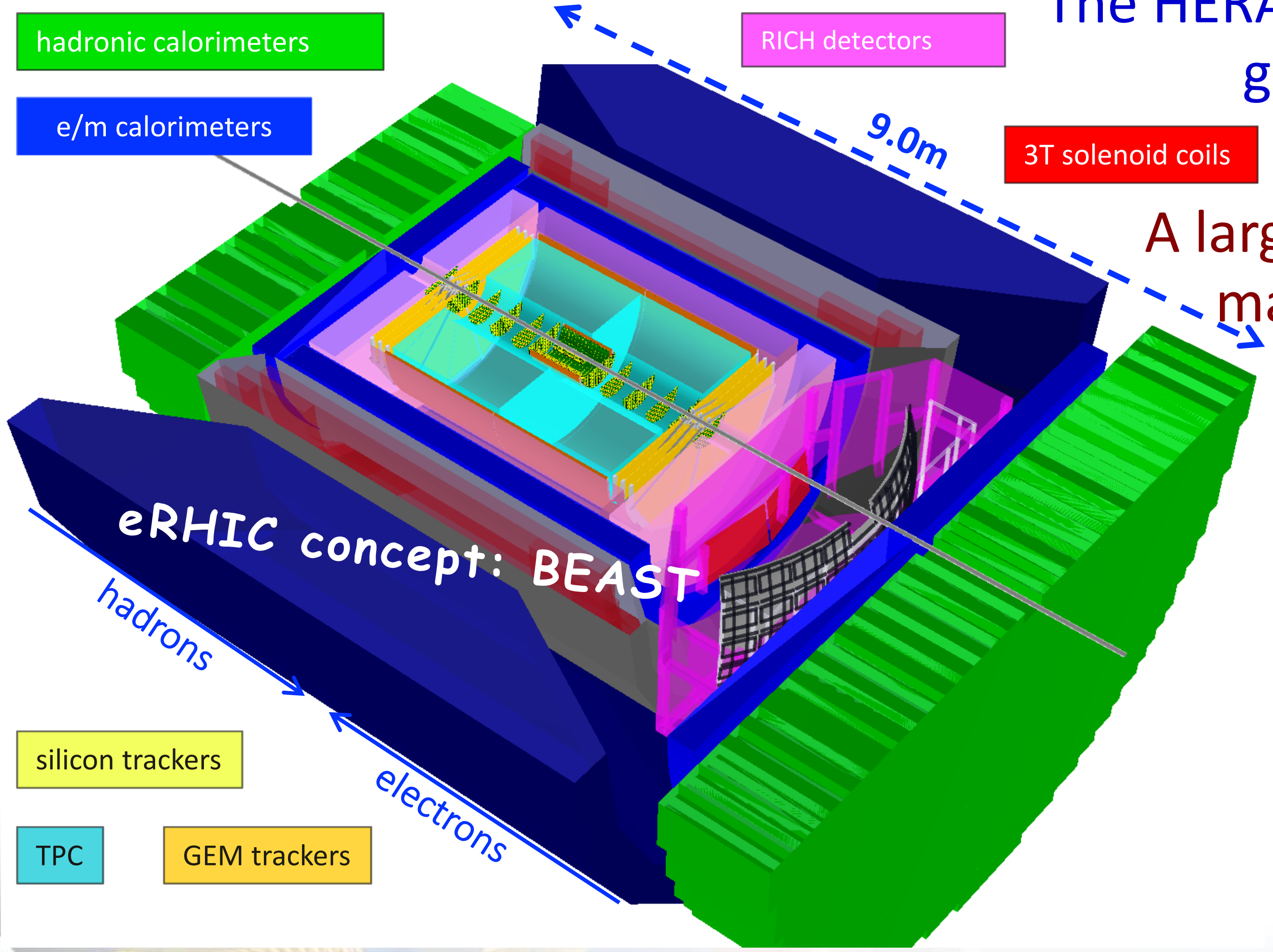
EIC PID needs are more demanding than your normal collider detector

EIC needs absolute particle numbers at high purity and low contamination

Ideal is to measure everything, everywhere.
~100% acceptance for the whole event!

BNL 25 mrad → eRHIC
JLab 50 mrad → JLEIC

For our EIC, why use a crossing angle?



The HERA and LHeC concepts have separation dipoles to get the beams apart… but dipoles generate synrad which can lead to backgrounds to avoid.

A large diameter Detector Integrated Dipole, creates magnetic “holes,” i.e. non-cylindrically-symmetric acceptance and momentum resolution.

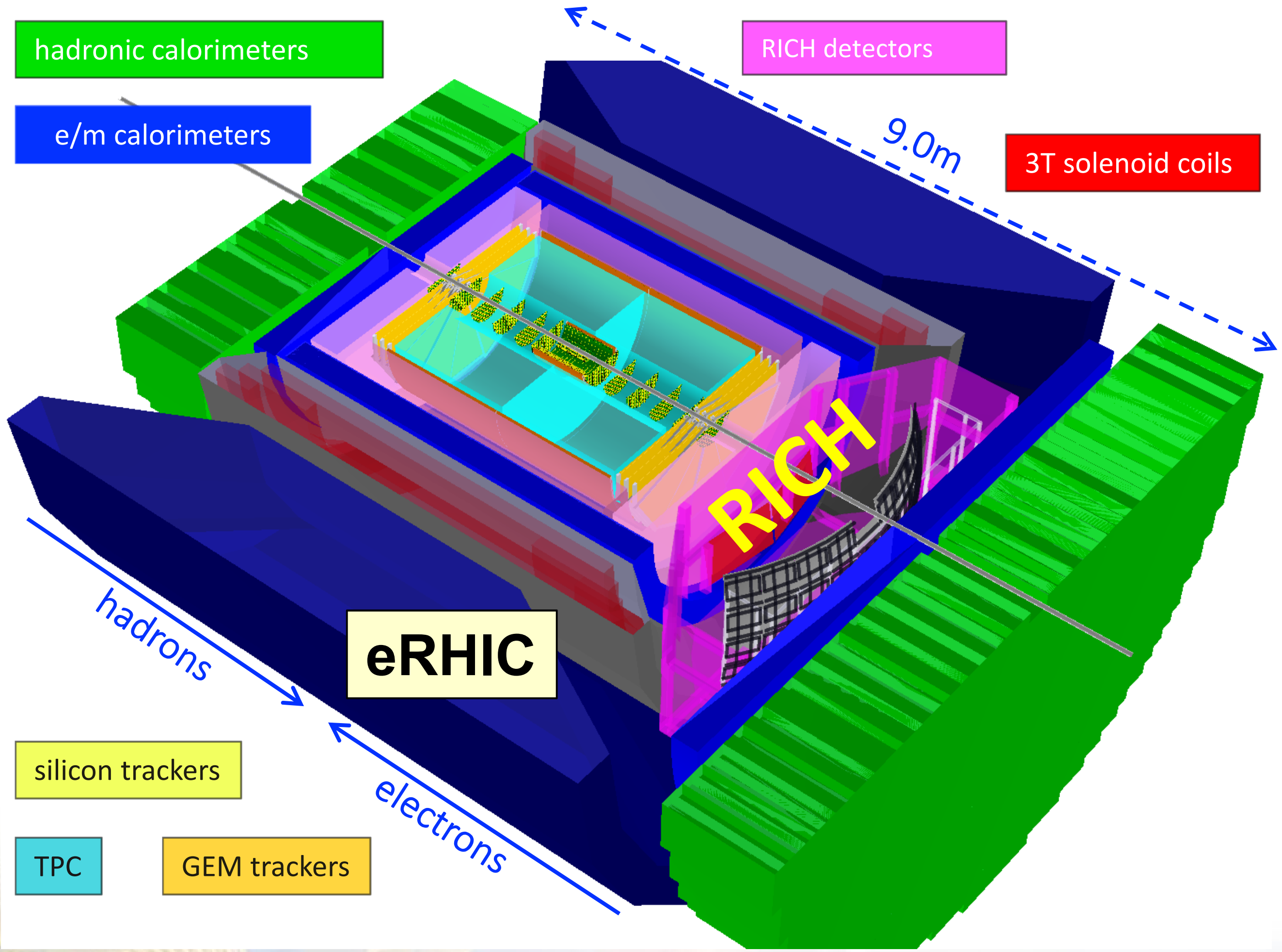
However with a large crossing angle, IR magnet designs get easier with rapid beam separation!

Crab Cavity luminosity recovery has not yet been demonstrated with hadron beams (where there is no natural damping).

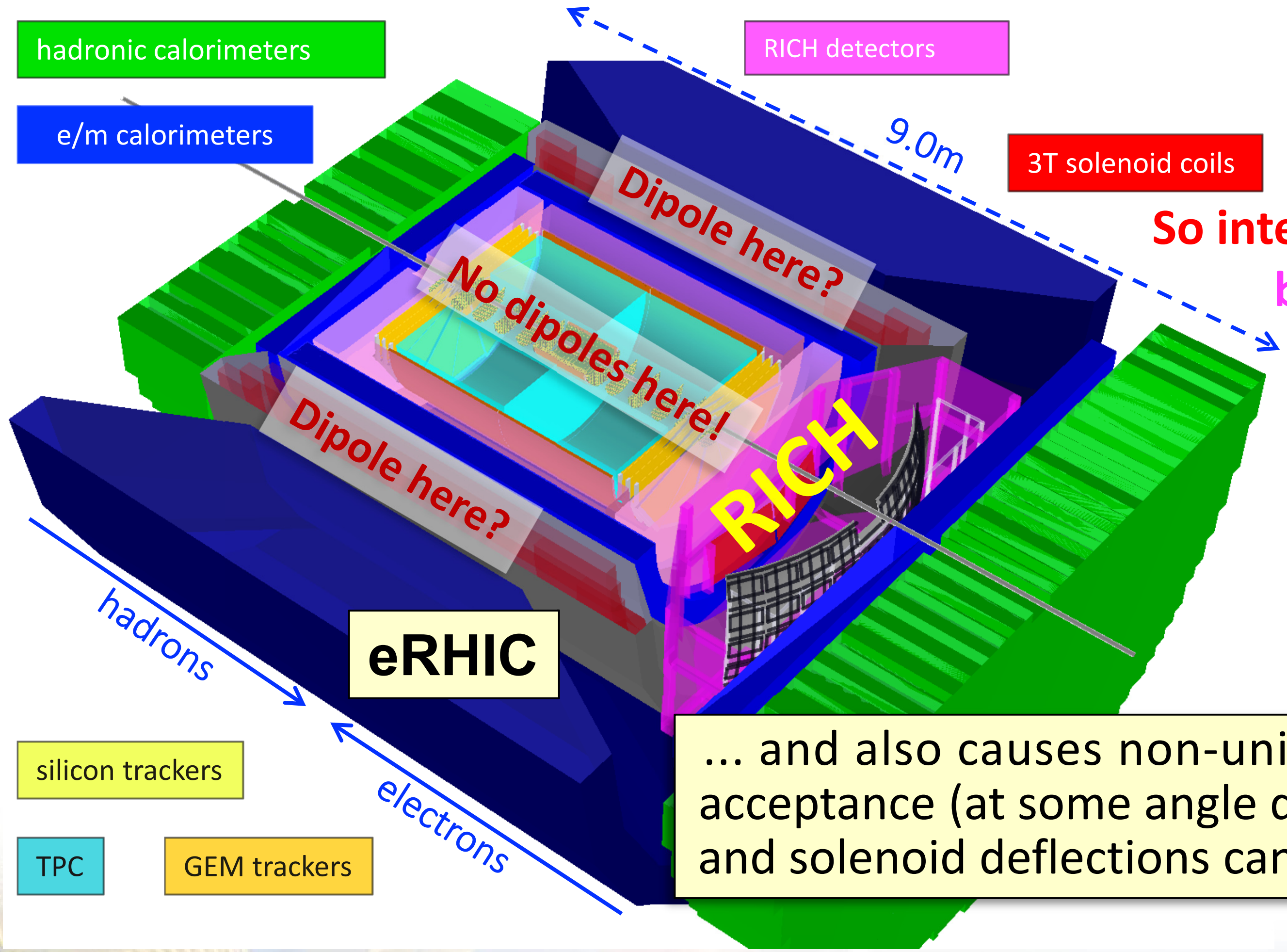
The EIC tie breaker is PID.

E.C. Aschenauer, A. Kiselev, B. Parker, and R. Petti, “Issues of Incompatibility of Dipole Beam Separation Schemes with EIC Physics,” EIC IR Group Design Report, BNL, Nov. 9, 2015, https://erhic.docdb.bnl.gov/cgi-bin/public/ShowDocument?docid=6.

So why don't we use separation dipoles?



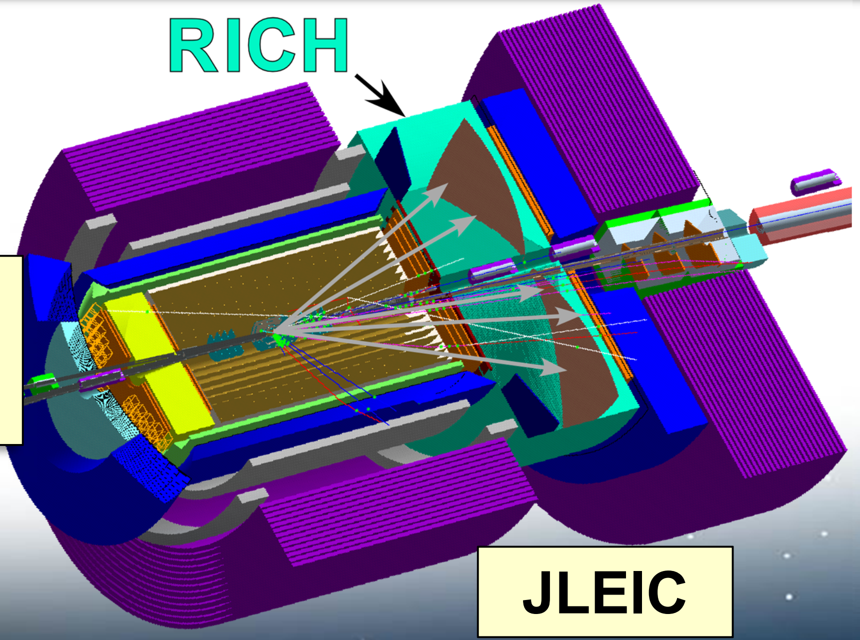
So why don't we use separation dipoles?



Placing small radius dipoles inside the detector looses important EIC physics.

So integrate dipole with solenoid,
but then cannot use a RICH!

Making a "projective" field for a RICH is not compatible with a large diameter dipole.

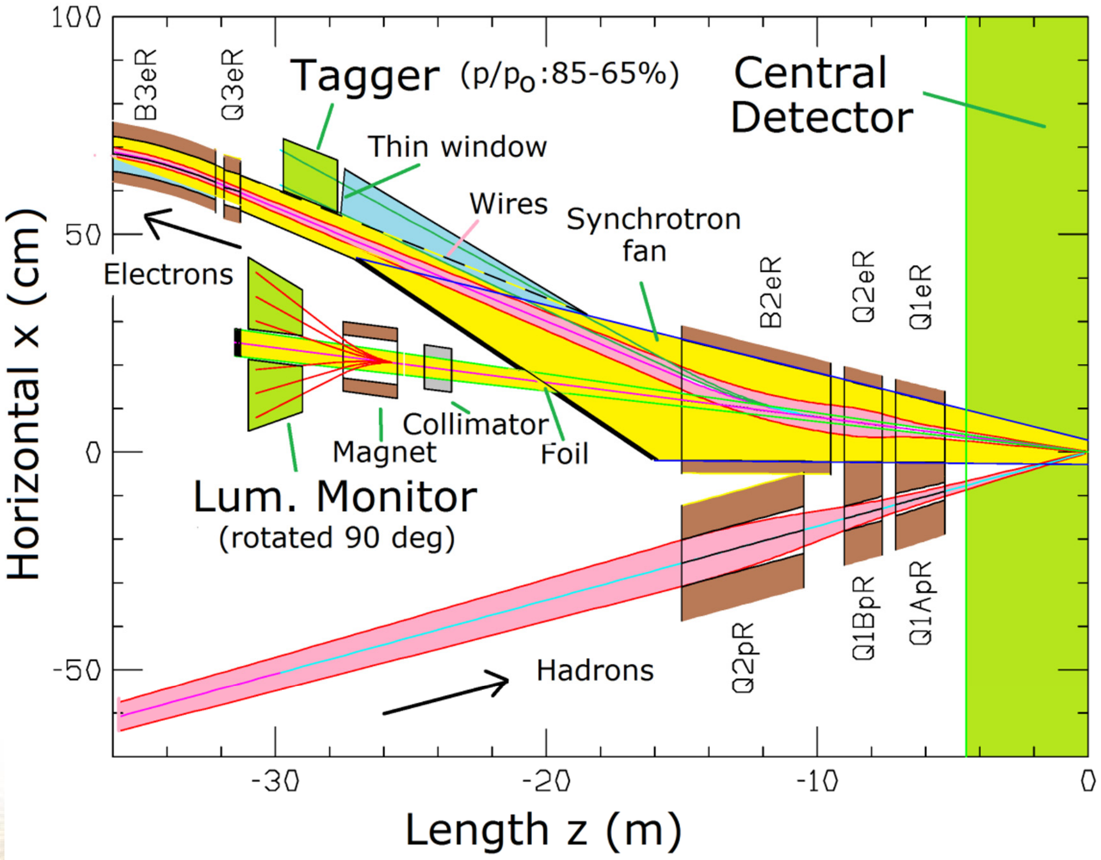


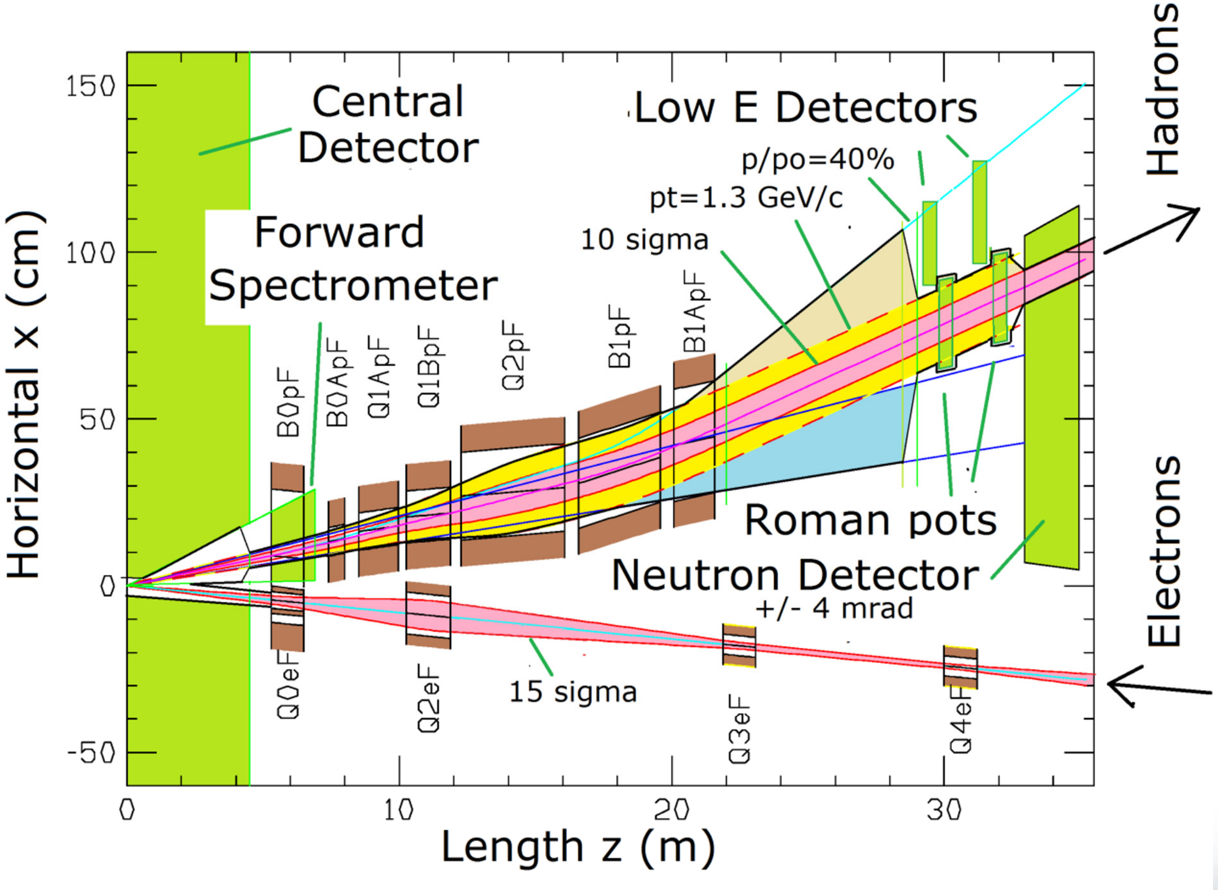
... and also causes non-uniform acceptance (at some angle dipole and solenoid deflections cancel!).

For RICH we want tracks aligned with magnetic field

IR Design Overview and Conventions.*

Rear Side **Detector Region** **Forward Side**





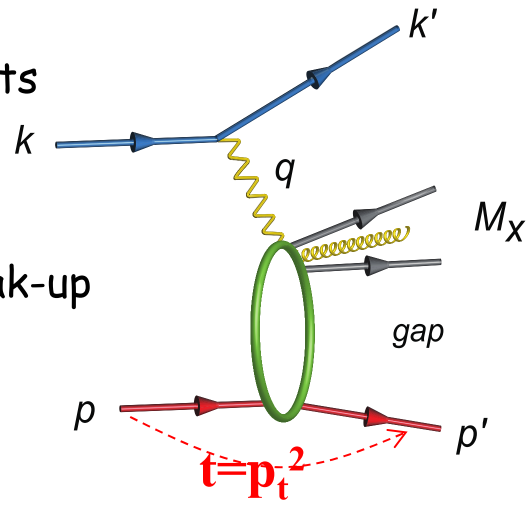
***The example shown is the BNL eRHIC IR configuration with 25 mrad total crossing angle.**

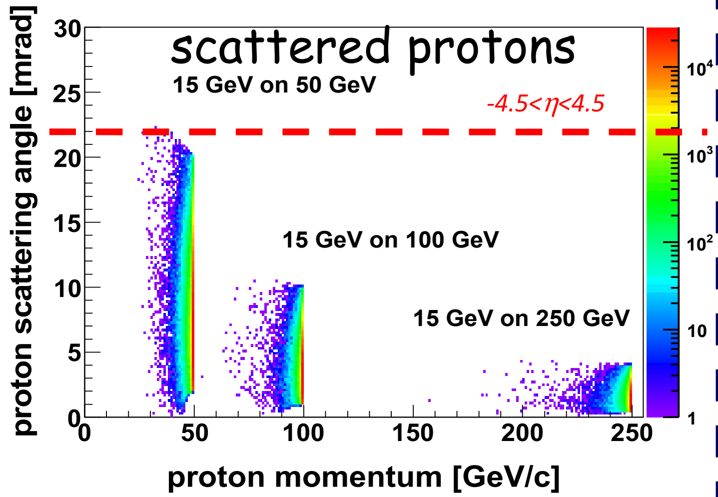
EIC Particle Detection Requirements.

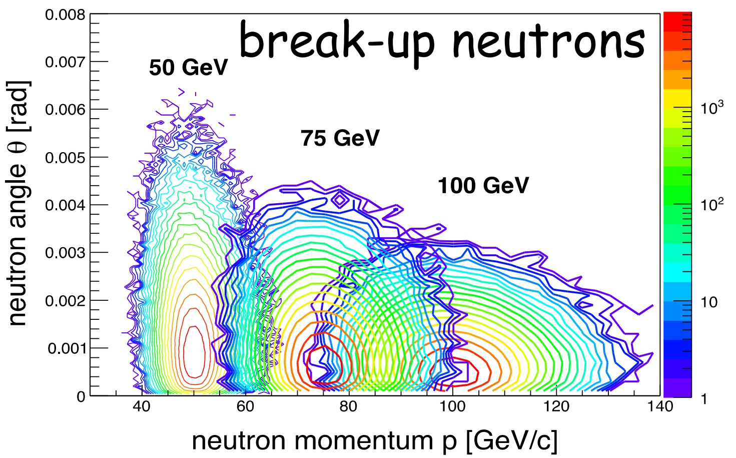
Exclusive Reactions in ep/eA:

- ❑ Exclusivity criteria:
  - ➢ eA: large rapidity coverage → rapidity gap events
    - ➢ HCal for 1<η<4.5
  - ➢ ep: reconstruction of all particles in event
    - ➢ wide coverage in t ($=p_t^2$) → Roman pots
- ❑ eA: large acceptance for neutrons from nucleus break-up
  - ➢ Zero Degree Calorimeter
    - ➢ veto nucleus breakup
    - ➢ determine impact parameter of collision

→ Poses stringent requirements on IR design

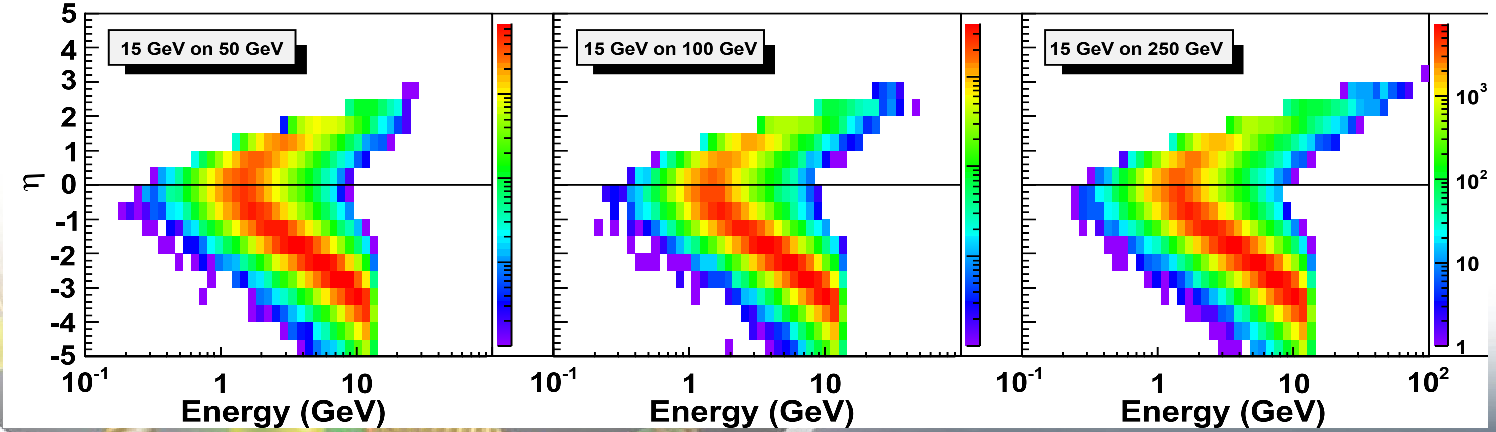






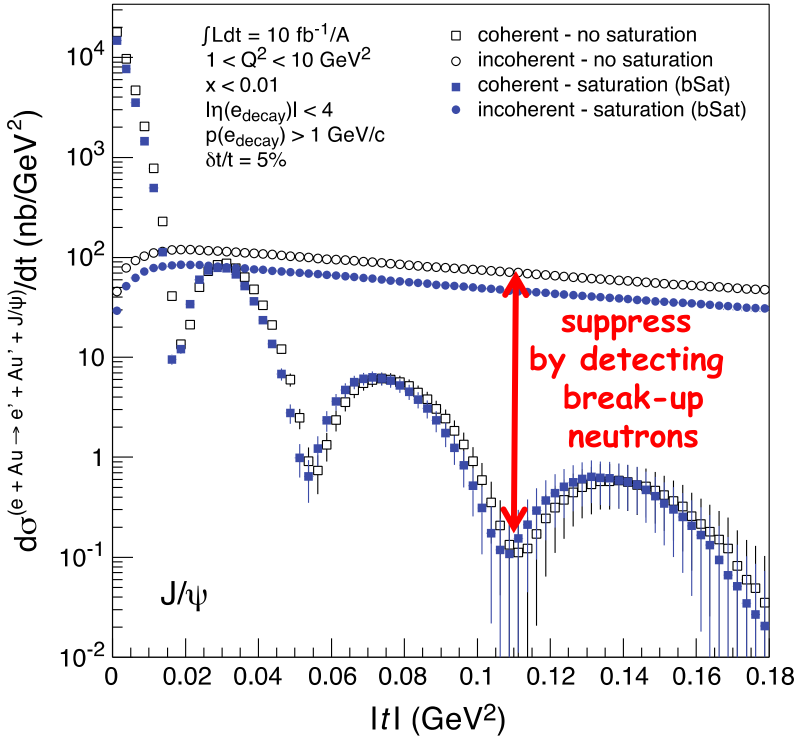
DVCS – photon kinematics:

Cuts: Q^2>1 GeV, 0.01<y<0.85

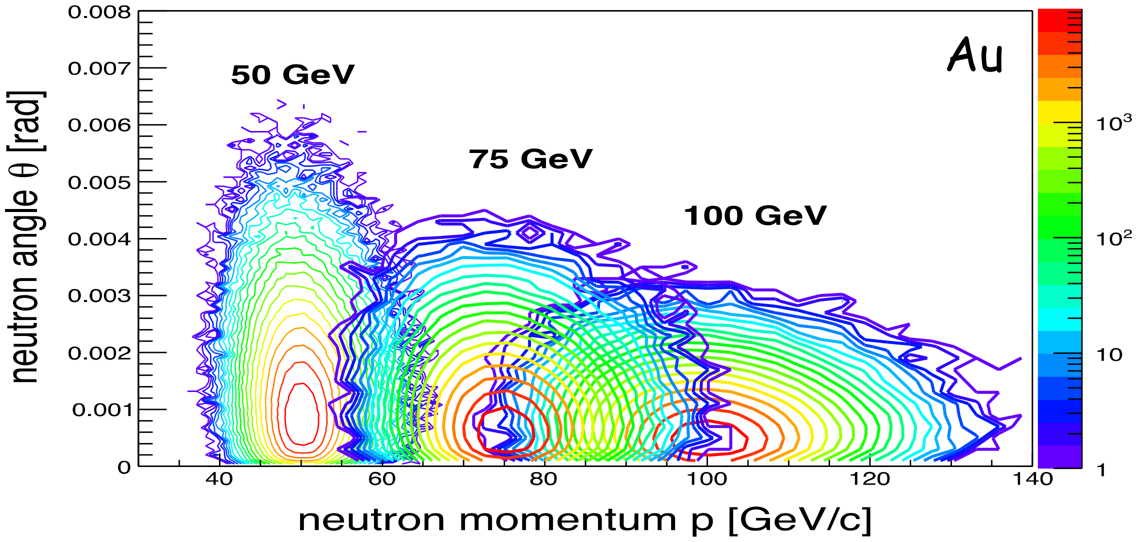


EIC Forward Scattered Neutrons.

Diffractive physics in eA
→ Measure spatial gluon distribution in nuclei
→ Reaction: $e + Au \rightarrow e' + Au' + J/\psi, \varphi, \rho$
→ Momentum transfer $t = |p_{Au} - p_{Au'}|^2$



Physics requires forward scattered nucleus needs to stay intact
→ Veto breakup through neutron detection

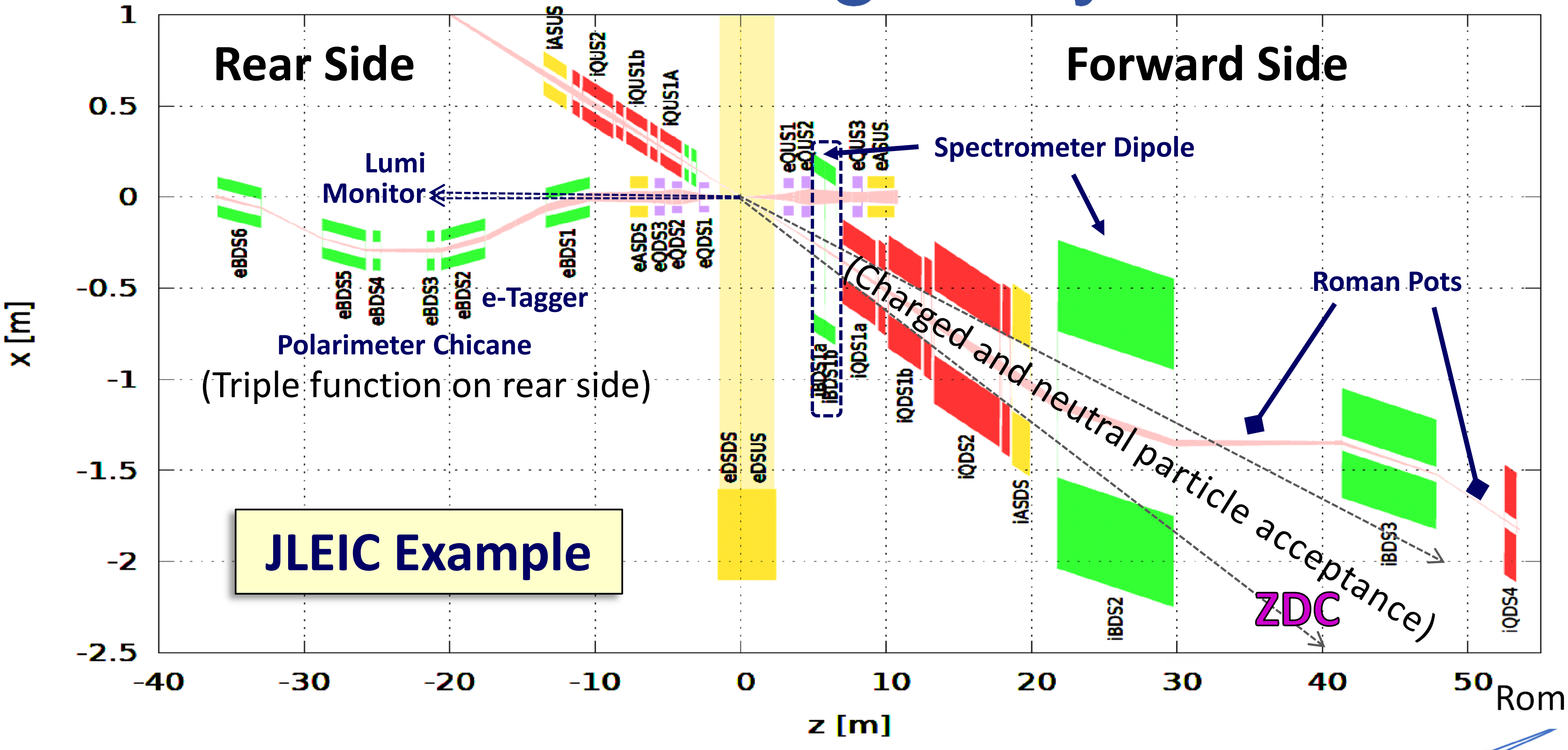


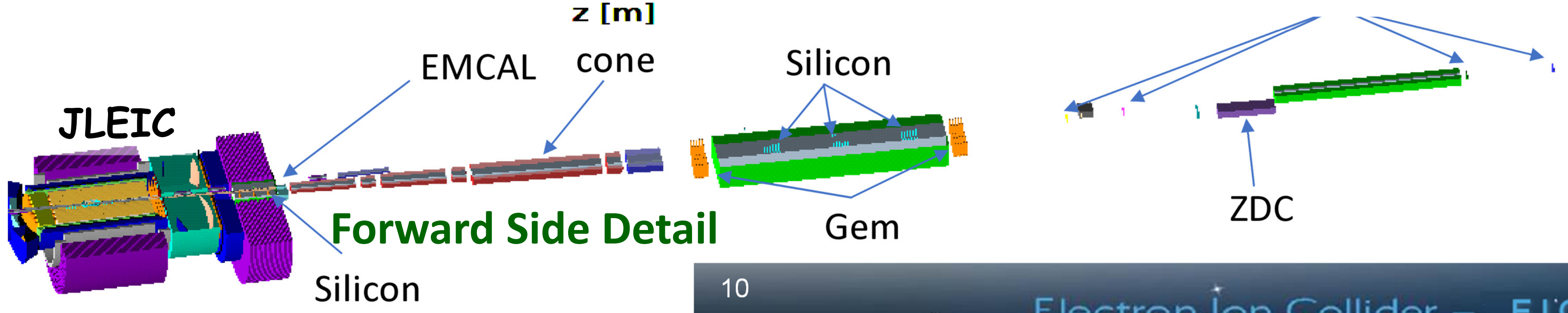
Requirement:
- Need at +/- 4 mrad beam element free cone before the zero degree calorimeter to detect the breakup neutrons
- No nuclear dependence for heavy A
- there neutrons are also critical to reconstruct collisions geometry

  L. Zheng, ECA, J-H. Lee arXiv:1407.8055

  → precision neutron energy reconstruction → transv. size of ZDC

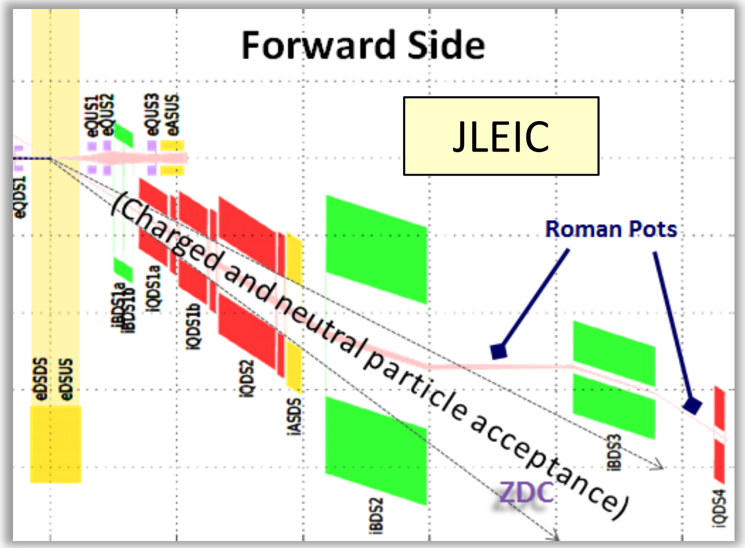
JLab EIC IR Design Layout Schematic.

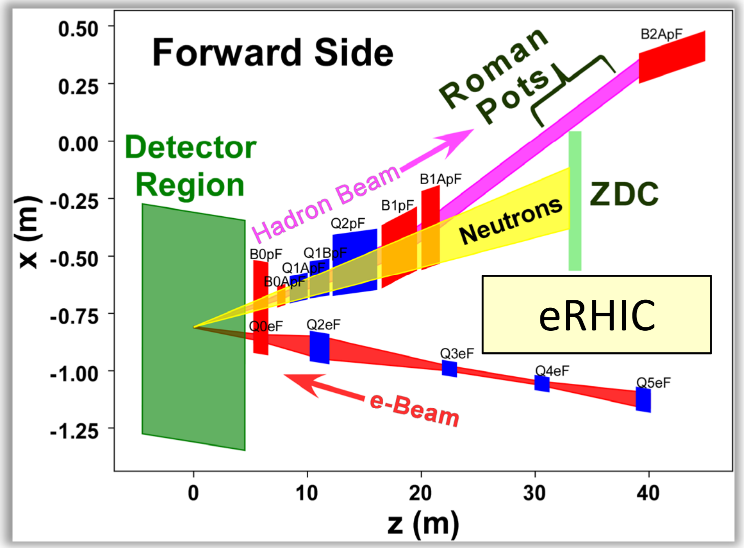




Detecting EIC Very Forward Scattered Particles.

The very forward scattered particles are critical to detect for elastic/diffractive physics and not detectable in main detector.





Note: Main detector acceptance starts at 35 mrad

Full p_T coverage of proton critical for physics

$p_T = p'_L \sin(\Theta)$ $p'_L > 97\%$ of p_{Beam}

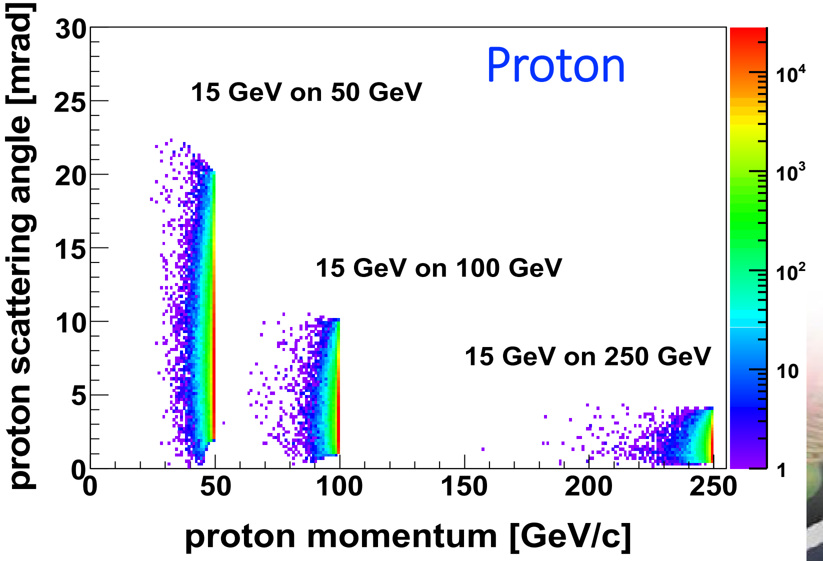
→ $0.65 < \Theta$ (mrad) < 10

→ 0.2 GeV $< p_t < 1.3$ GeV

→ small p_T-acceptance limited by beam divergence, emittance and beta-function

→ rule of thumb keep 10σ between Roman Pot and circulating beam

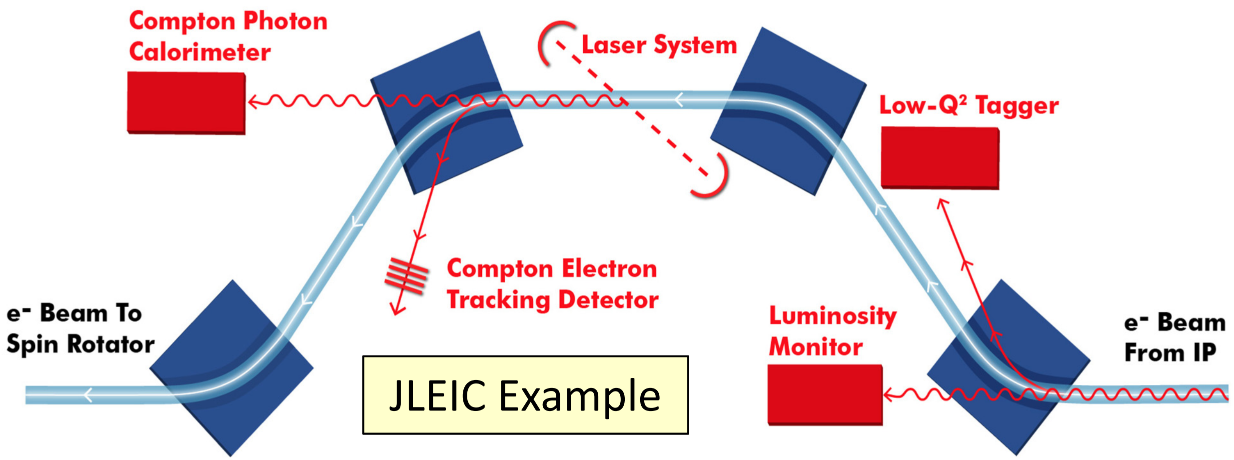
→ large p_T-acceptance limited by magnet apertures



EIC adopts a solution that makes use of both:

- An instrumented spectrometer dipole right next to the main detector,
- And Roman Pots that approach as close as possible to the core of the circulating hadron beam but are far from the main detector region.

Precision Luminosity & Polarization Measurement.



- Jlab EIC layout (above) combines the functionality of lumi detector, the lepton polarimeter and rear electron tagging in one chicane.
- BNL EIC uses a dogleg beam bump for the lumi detector and rear electron tagging and does lepton polarimetry in the same region where hadron polarimetry is located.

EICUG IR / Luminosity Working Group

- ➢ interface between accelerator / IR design and the physics requirements
- ➢ proper implementation of the physics program in particular for forward / backward detection

Conveners:

- Accelerator / IR: **C. Montag (BNL), V. Morozov (JLab)**
- Physics: **C. Hyde (ODU), A, Kiselev (BNL)**

EICUG Working Group on Polarimetry

➢ plan / develop the optimal methods and techniques for measuring electron/ion beam polarization with high precision

Conveners:

- **E.-C. Aschenauer (BNL), D. Gaskell (JLab)**

Workgroups explore common issues

Lepton Beam Luminosity Measurement

Luminosity Detector

Concept:

Use Bremsstrahlung ep → epγ as reference cross section

- different methods:
- Bethe Heitler, QED Compton, Pair Production
- Hera: reached 1-2% systematic uncertainty

Goals for Luminosity Measurement:

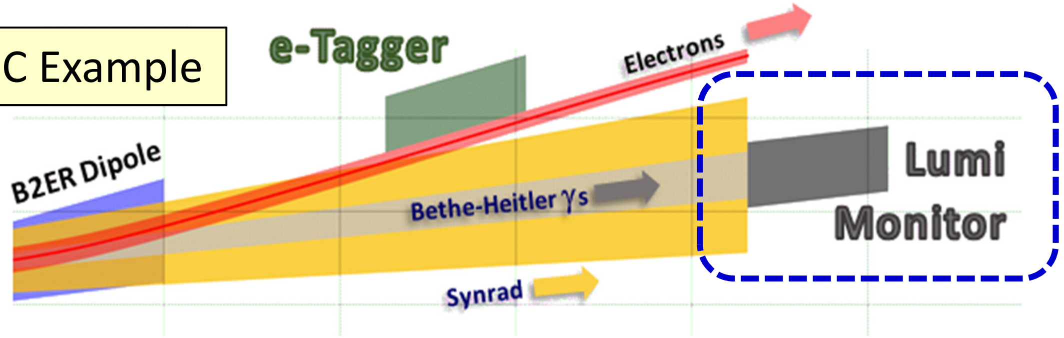
- Integrated luminosity with precision δL/L< 1%
- Measurement of relative luminosity:
  physics-asymmetry/10 → ~10^{-4} – 10^{-5}

EIC challenges:

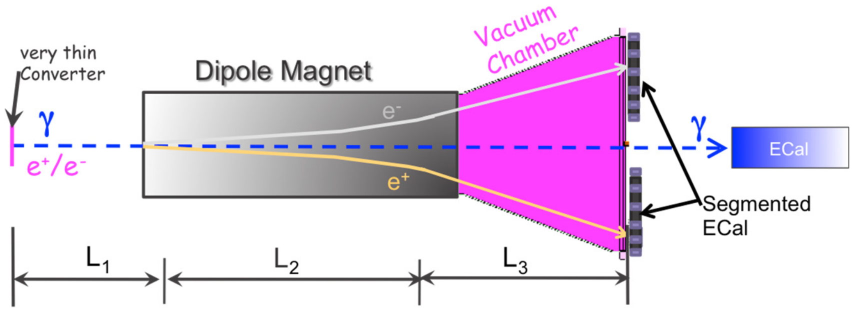
- with 10^{33} $cm^{-2}s^{-1}$ one gets on average 23 bremsstrahlungs photons/bunch for proton beam
  → A-beam Z^2-dependence
- Need more sophisticated solution
- BH photon cone < 0.03 mrad
  → acceptance completely dominated by lepton beam size

Workgroup explores common issues

eRHIC Example



Lumi Monitor Schematic



Pair Spectrometer

low rate

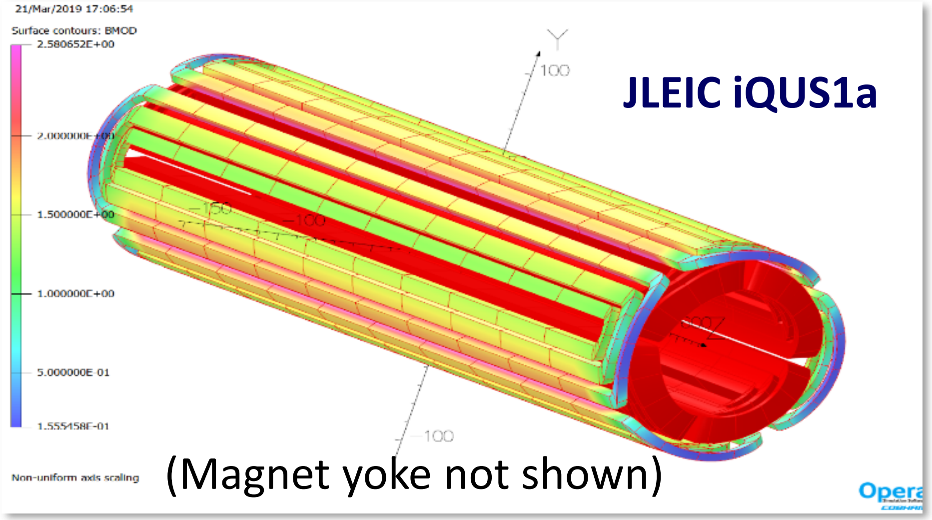
- ➢ High precision measurement for physics analysis
- ➢ The calorimeters are outside of the primary synchrotron radiation fan

Zero Degree Photon Calorimeter

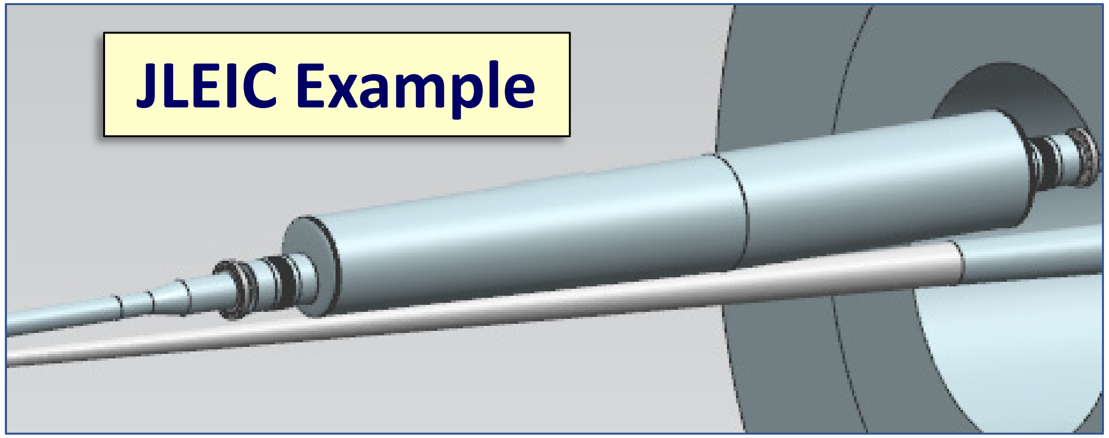
high rate

- ➢ Fast feedback for machine tuning
- ➢ measured energy proportional to # photons
- ➢ subject to synchrotron radiation

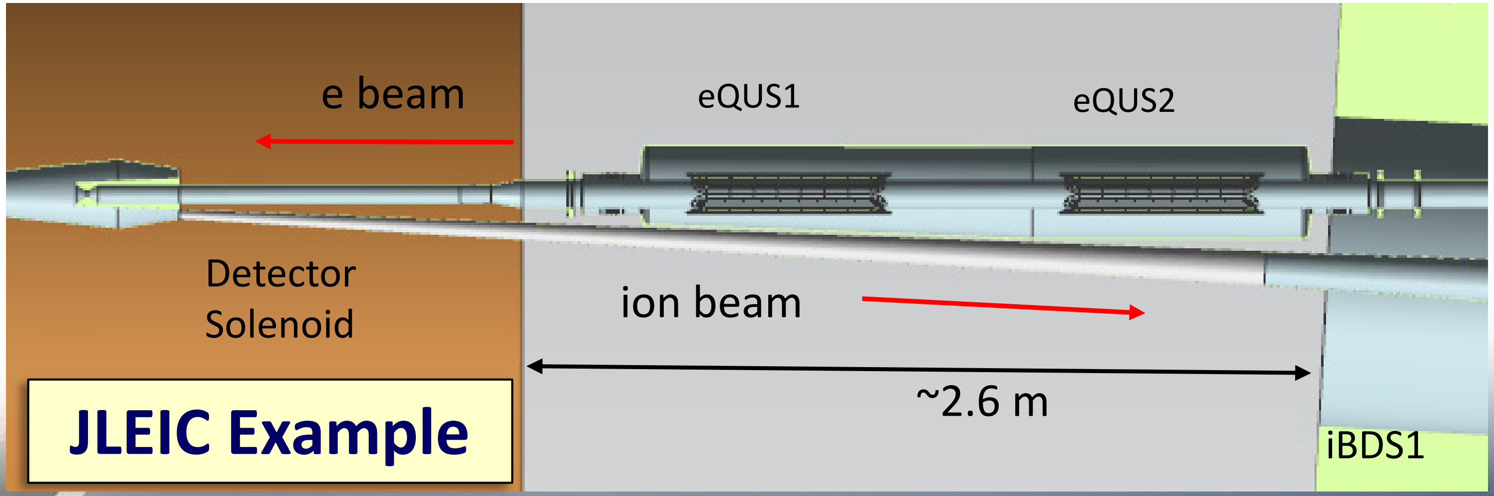
JLab EIC Magnets and Component Integration.



EIC IR designs require production of many new challenging magnets (fortunately NbTi seems to be ok and Nb_3Sn may not be needed).

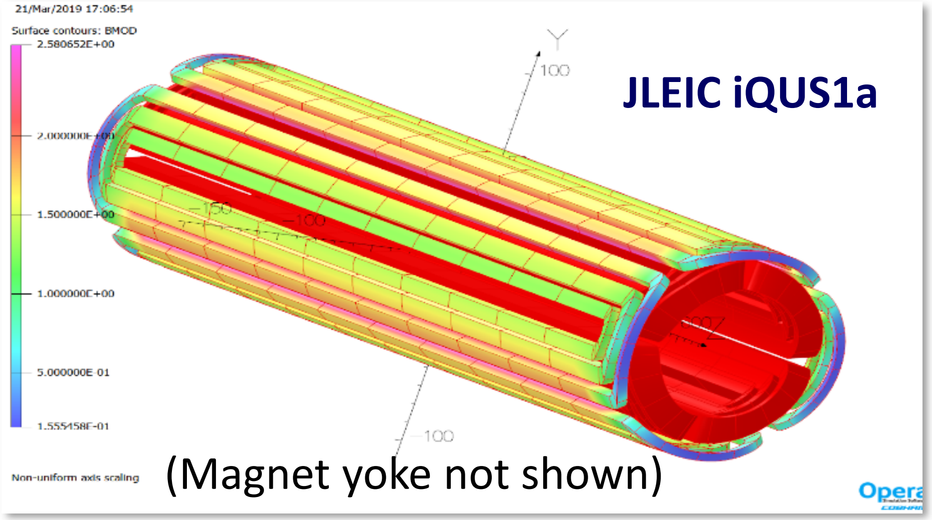


But there are still quite significant engineering challenges in order to fit everything in very restricted available space!



Many hadron superconducting magnets have large apertures, high coil peak fields and could have significant external fields.

JLab EIC Magnets and Component Integration.



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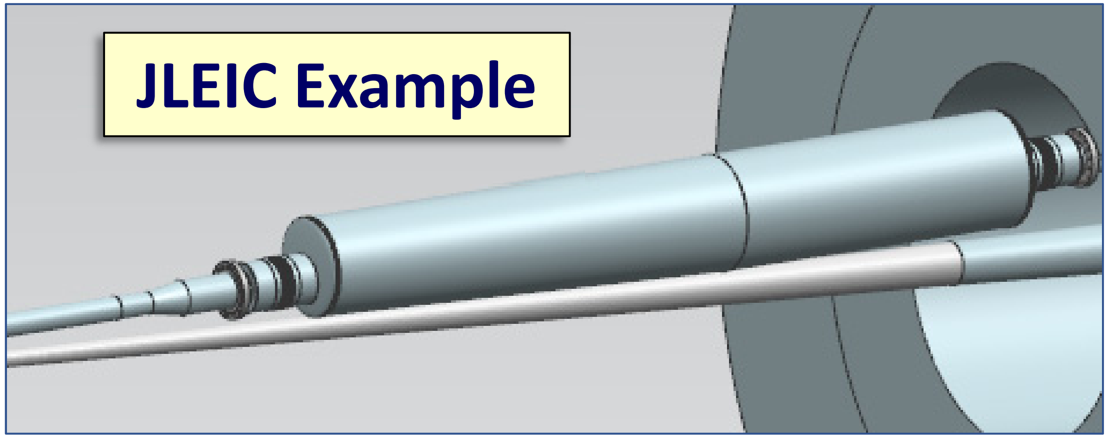
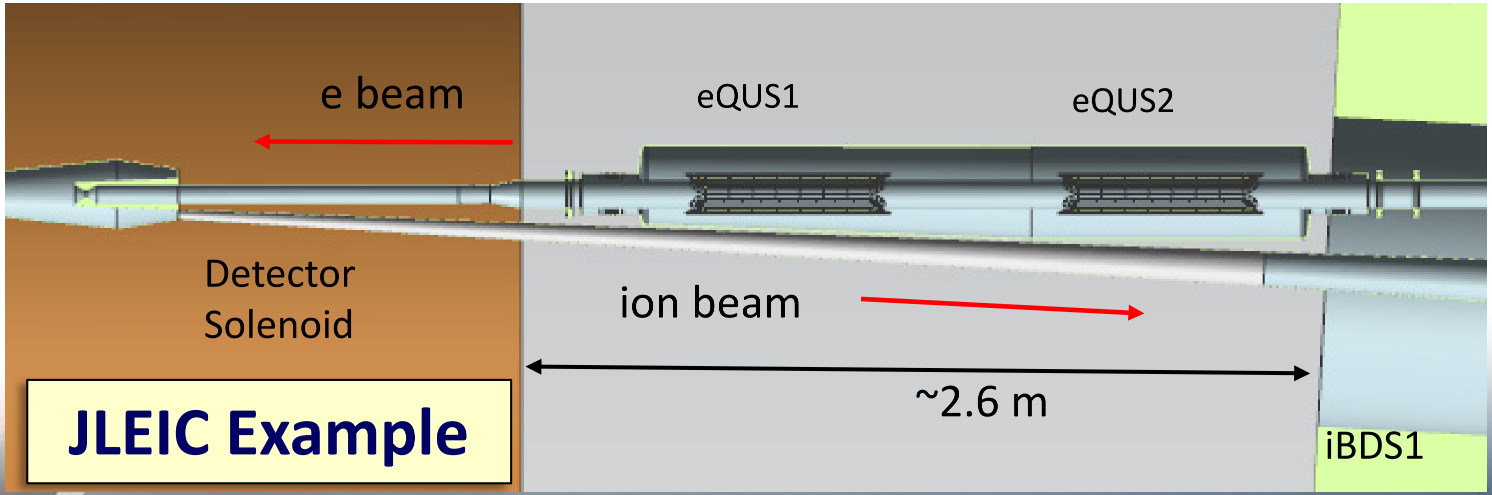
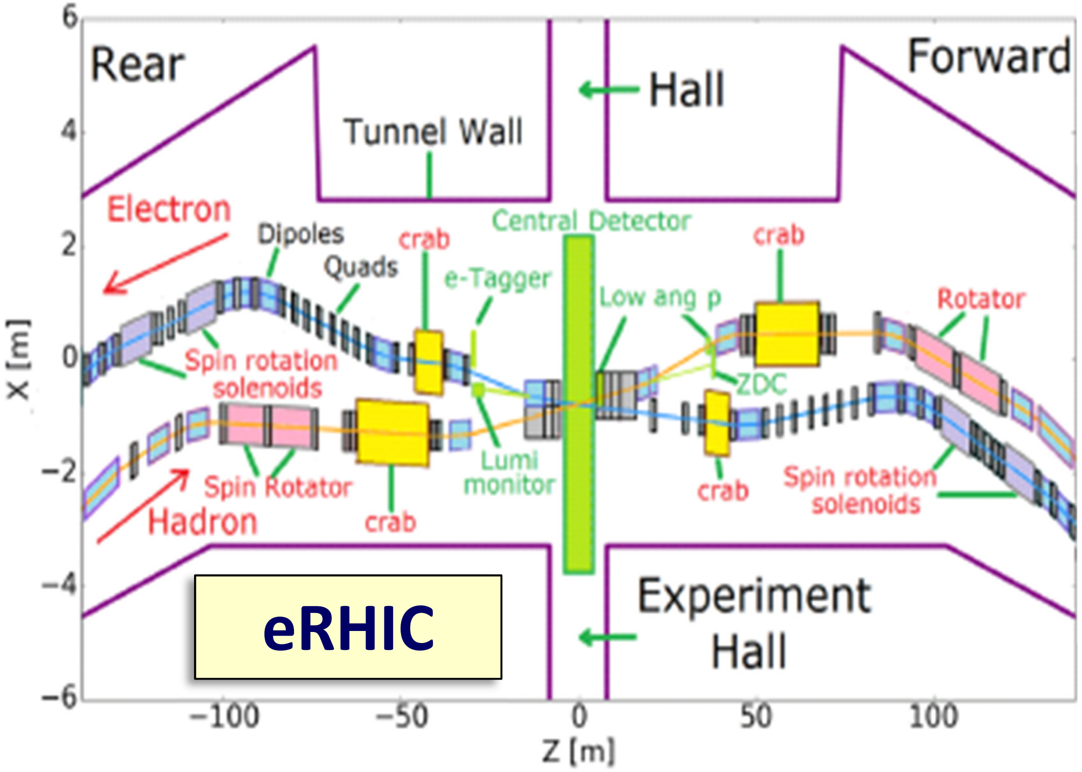


But there are still quite significant engineering challenges in order to fit everything in very restricted available space!

A very tough challenge is to always be sure to shield the electron beam from the quite strong hadron magnet fields!

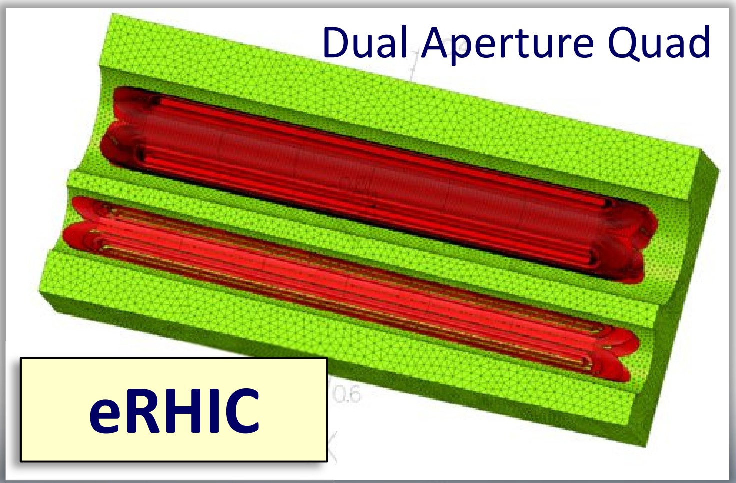


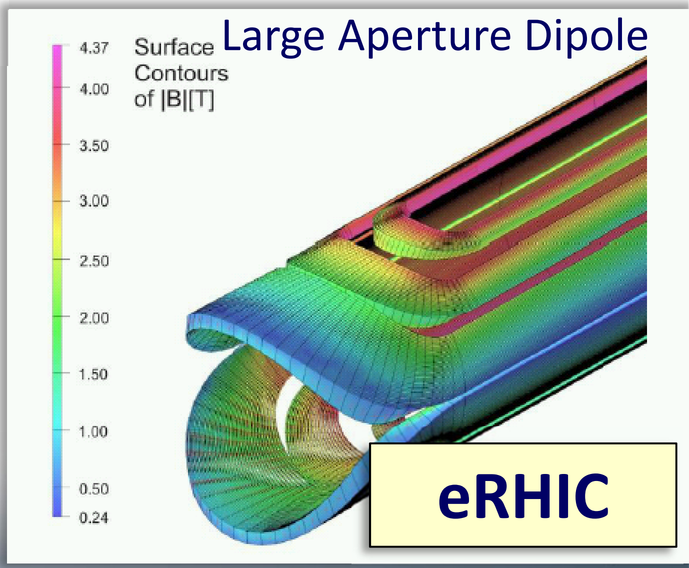
BNL EIC Magnets and Component Integration.



EIC IR designs require production of many new challenging magnets (fortunately NbTi seems to be ok and Nb_3Sn may not be needed).

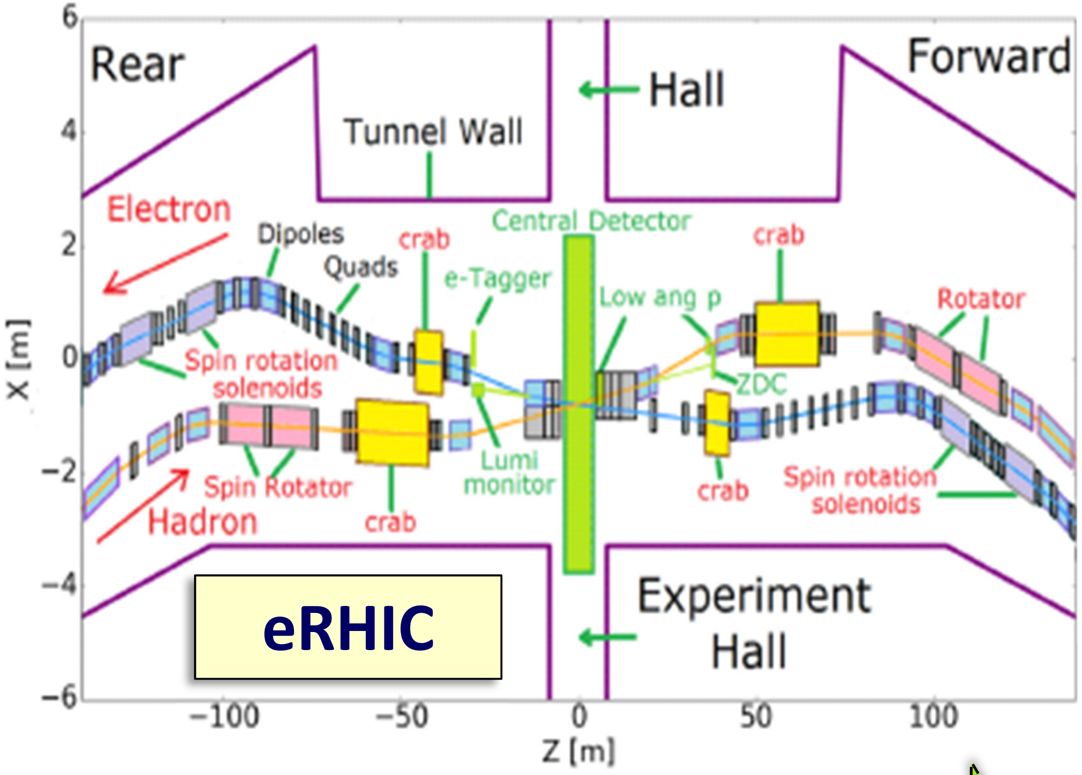
To pass synrad cleanly through rear side electron magnets, we make use of <u>tapered constant gradient quadrupole coils</u>!





Good, we fit in the existing RHIC tunnel!

BNL EIC Magnets and Component Integration.



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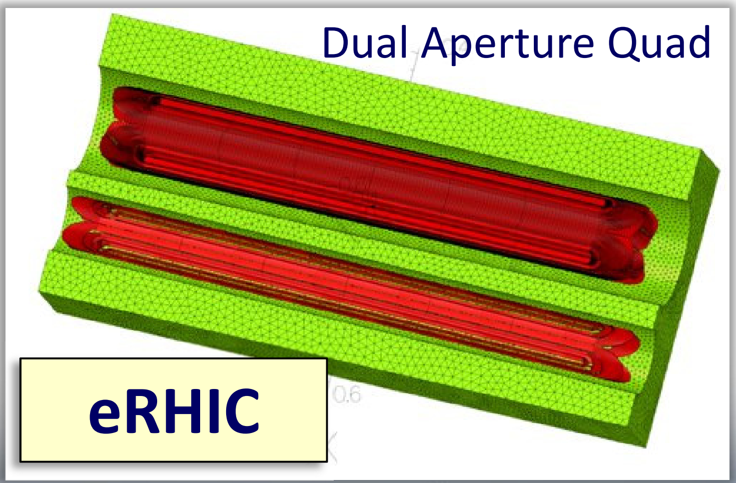
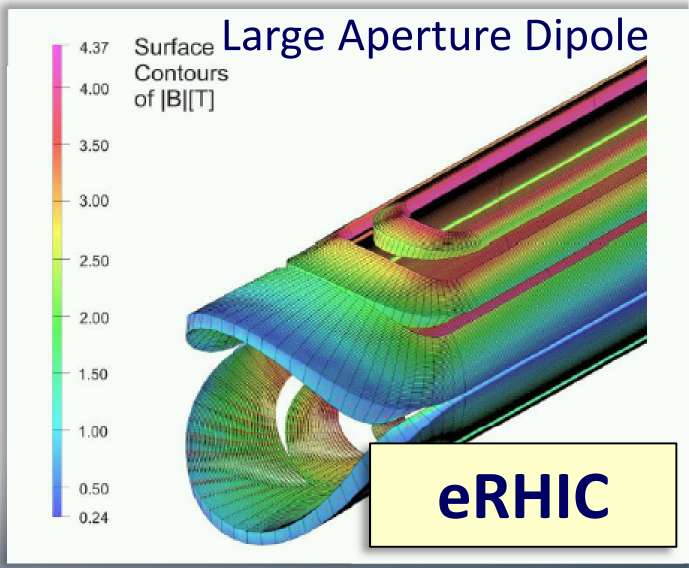
Dual Aperture Magnets

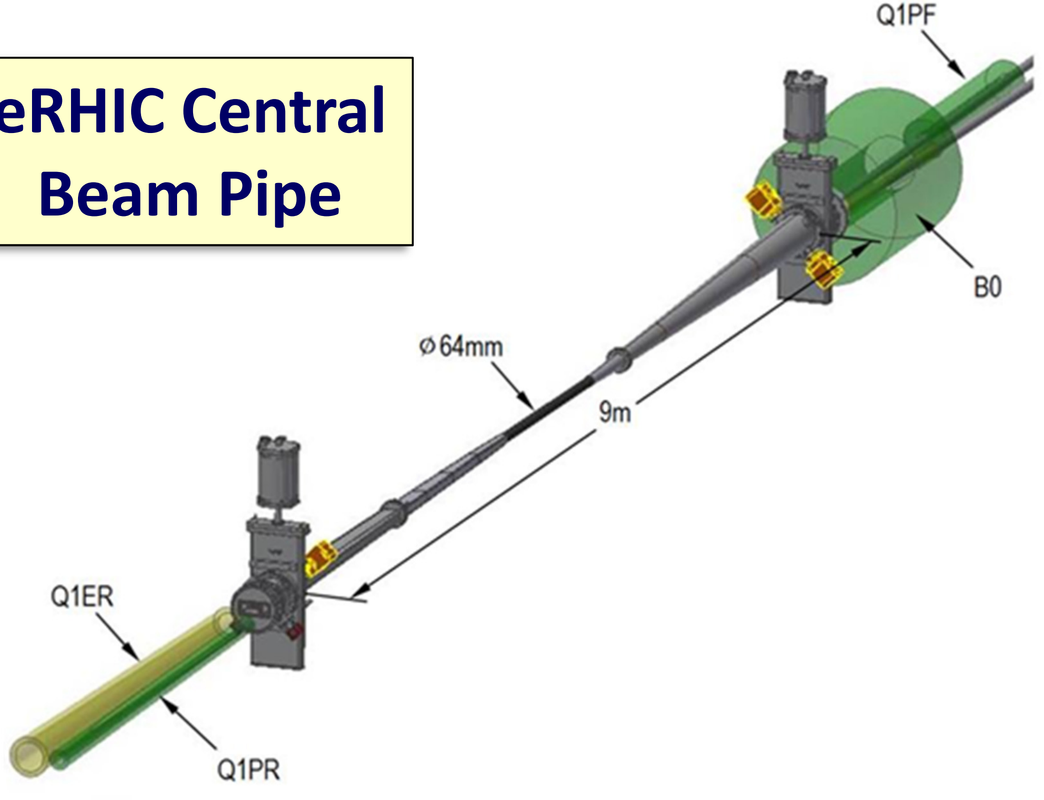


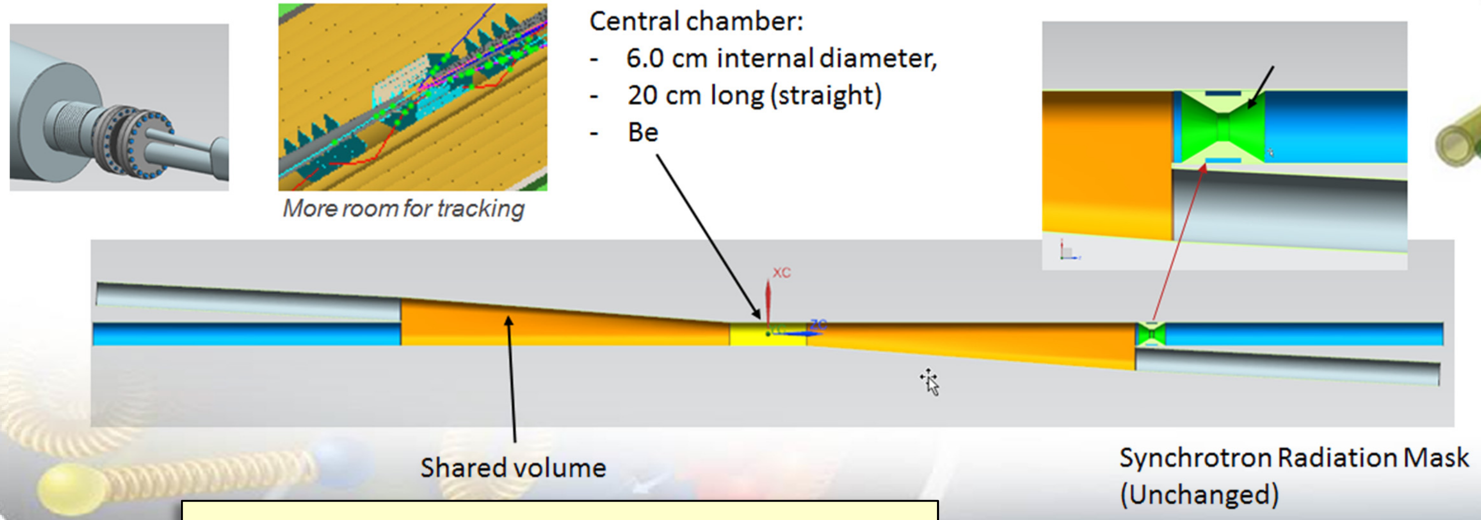


Detector Beam Pipes for an EIC Experiment.

- **Small diameter, thin walled Be in center region.**
- **Smooth tapers and transitions to limit energy deposited by beam in trapped modes (wakes).**
- **Need to provide beam tube cooling and for any Higher Order Mode (HOM) damping absorbers.**
- **Need to provide adequate vacuum pumping.**
- **eRHIC / Beast plans include *in situ* bakeout.**

eRHIC Central Beam Pipe





JLEIC Central Beam Pipe

<u>Central beam pipe requires meticulous optimization.</u>

Wrap Up: EIC Collaboration Acknowledgment.

Within the limited time of this presentation, we have covered the main highlights of EIC MDI issues.

The work presented here and a greater body of work that did not fit the allotted time, was performed collaboratively by people in the EIC IR working groups and EIC user group teams, who are too numerous to individually credit here.

The authors of this paper wish to thank everyone for their dedication to our common vision of realizing an EIC as the next high priority US Nuclear Physics project.

Thank you for your attention.